This assignment will not be graded and is only for practice.

Level 1

1) Let $f(x, y) = x\log(y)$ and $Hf(x, y)$ denote the Hessian matrix of f at (x, y). Which of the following options are correct? ***1 point***

- [ ] Rank of $Hf(0, 1)$ is 2.
- [ ] Rank of $Hf(0, 1)$ is 1.
- [ ] $(0, 1)$ is a local minimum.
- [ ] $(0, 1)$ is a saddle point.

No, the answer is incorrect.
Score: 0

Accepted Answers:

Rank of $Hf(0, 1)$ is 2.
$(0, 1)$ is a saddle point.

2) Choose the correct statements about $f(x, y) = x^2 + axy + by^2$. ***1 point***

- [ ] $(0, 0)$ is a saddle point of f iff $4b - a^2 < 0$.
- [ ] $(0, 0)$ is a saddle point of f iff $4b - a^2 > 0$.
- [ ] There are infinitely many possible pairs (a, b) for which f has local minimum at $(0, 0)$.
- [ ] If $a = b = 1$ then f has a local maximum at $(0, 0)$.

No, the answer is incorrect.
Score: 0

Accepted Answers:

$(0, 0)$ is a saddle point of f iff $4b - a^2 < 0$.
There are infinitely many possible pairs (a, b) for which f has local minimum at $(0, 0)$.

3) Consider the following function **1 point**

$$f(x,y) = x^2 + xy + y^2 + 2x - 2y + 5$$

Which of the following option(s) is (are) true?

- [ ] $(-2,2)$ is a critical point of f.
- [ ] $(2,-2)$ is a critical point of f.
- [ ] Determinant of the Hessian matrix at point (x,y) is 3.
- [ ] f has a local minimum at point $(-2,2)$.
- [ ] $(2,-2)$ is a saddle point of f.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$(-2,2)$ is a critical point of f.
Determinant of the Hessian matrix at point (x,y) is 3.
f has a local minimum at point $(-2,2)$.

4) For what value of k is the Hessian test inconclusive for the function $f(x,y) = x^3 + kxy + y^3$ at $(0,0)$?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 0

1 point

Level 2

5) Choose the correct statements about $f(x,y) = 2x^3 - 3x^2y - 12x^2 - 3y^2$. **1 point**

- [ ] f has 3 critical points.

- [ ] $(0, 0)$ is a local maximum.
- [ ] $(0, 0)$ is a saddle point.
- [ ] $(-4, -8)$ is a saddle point.
- [ ] $(-4, -8)$ is a local minimum.
- [ ] f has a global minimum.

No, the answer is incorrect.

Score: 0

Accepted Answers:

f has 3 critical points.
$(0, 0)$ is a local maximum.
$(-4, -8)$ is a saddle point.

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6) Choose the correct statements about $f(x, y) = x^8 - 2x^4y + y^3 - y$. **_1 point_**

- [ ] The Hessian matrix of f at point (x, y) is $\begin{bmatrix} 56x^6 - 24x^2y & -8x^3 \\ -8x^3 & 6y \end{bmatrix}$.
- [ ] The Hessian matrix of f at point (x, y) is $\begin{bmatrix} 56x^6 - 24x^2y & 8x^3 \\ 8x^3 & 6y \end{bmatrix}$.
- [ ] f has four critical points.
- [ ] $(1, 1)$ is a point of local minimum of f.
- [ ] $(1, 1)$ is a point of local maximum of f.
- [ ] Second derivative test is inconclusive for the point $(0, \frac{1}{\sqrt{3}})$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The Hessian matrix of f at point (x, y) is $\begin{bmatrix} 56x^6 - 24x^2y & -8x^3 \\ -8x^3 & 6y \end{bmatrix}$.
f has four critical points.
$(1, 1)$ is a point of local minimum of f.
Second derivative test is inconclusive for the point $(0, \frac{1}{\sqrt{3}})$.

7) If M denotes the local maximum value of the function $F(a,b) = \int_a^b 3 - 2x - x^2 \, dx$ over all possible values of a and b then find the value of $3M$. Note that the function has only one local maximum and you have to find that.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 32

1 point

8) Consider a function $f(x,y)$. Let v_1, v_2, v_3 be the critical points of the function f and H_k be the Hessian matrix of the function at the point v_k.Then which of the following options is(are) true? ***1 point***

- [ ] If $H_1 = \begin{bmatrix} 1 & 0 \\ -3 & 2 \end{bmatrix}$, then v_1 is a saddle point.
- [ ] If $H_2 = \begin{bmatrix} 1 & 1 \\ 3 & 2 \end{bmatrix}$, then v_2 is a saddle point.
- [ ] If $H_3 = \begin{bmatrix} 3 & 1 \\ -3 & 1 \end{bmatrix}$, then v_3 is a point of local extremum.
- [ ] If $H_1 = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$, then v_1 is a point of local minimum.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If $H_2 = \begin{bmatrix} 1 & 1 \\ 3 & 2 \end{bmatrix}$, then v_2 is a saddle point.

If $H_3 = \begin{bmatrix} 3 & 1 \\ -3 & 1 \end{bmatrix}$, then v_3 is a point of local extremum.

If $H_1 = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$, then v_1 is a point of local minimum.

Your score is: 0/8